

Johnson Schemes: Permutation Molien Formula Proof and Exact Verification

Executable companion note

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Abstract

For the permutation representation on the k -subsets of $[v]$, we prove the cycle-factor and orbit formulas, derive the fixed-subset formula, treat the middle-dimensional complementation coset, and verify all coefficients through total degree three in three representative finite cases. The computation uses the actual permutation matrices and does not assume the desired cycle formula.

1 Master identity

Let a finite group G act on a finite set X , let $V = \mathbb{C}[X]$, and let $c_\ell(g)$ count the ℓ -cycles of g on X . Then

$$\mathcal{S}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \prod_{\ell \geq 1} (1 - t_i^\ell)^{-c_\ell(g)}. \quad (1)$$

Indeed, the eigenvalues of an ℓ -cycle are all the ℓ th roots of unity, so $\det(I - tP_\ell) = 1 - t^\ell$. Multiplication over disjoint cycles and the finite-group Molien average prove (1).

For a partition $\tau = (\tau_1, \dots, \tau_s)$, $\text{Sym}^{\tau_j}(V)$ is the permutation module on multisets of size τ_j in X . Burnside's lemma therefore gives the independent interpretation

$$[m_\tau] \mathcal{S}_{G,X} = \#(G \backslash (\text{Sym}^{\tau_1} X \times \dots \times \text{Sym}^{\tau_s} X)). \quad (2)$$

2 Johnson fixed points and complementation

Put $X = \binom{[v]}{k}$. If $\sigma \in S_v$ has a_j cycles of length j , then a j -cycle splits under σ^r into $\gcd(j, r)$ cycles of length $j/\gcd(j, r)$. A fixed subset is exactly a union of these cycles; hence

$$F_r^J(\sigma) = [u^k] \prod_{j \geq 1} \left(1 + u^{j/\gcd(j,r)}\right)^{a_j \gcd(j,r)}. \quad (3)$$

The vertex-cycle counts follow from

$$c_\ell(\sigma) = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d^J(\sigma). \quad (4)$$

When $v = 2k$ and $k > 1$, complementation $T(A) = A^c$ commutes with S_{2k} and supplies the additional coset in the full Johnson-scheme automorphism group. For odd r , a subset fixed by $\sigma^r T$

must alternate membership around every cycle of σ^r . Thus

$$F_r(\sigma T) = \begin{cases} F_r^J(\sigma), & r \text{ even,} \\ 2^{c(\sigma^r)}, & r \text{ odd and every cycle of } \sigma^r \text{ is even,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Every alternating choice has size exactly half of v , so no further size constraint is missing. Equations (1)–(5) prove the claimed formula, including the middle case.

3 Verification design

For each group element, the code explicitly builds its $|X| \times |X|$ permutation matrix P . It computes $h_d(P) = \text{tr}(\text{Sym}^d P)$ from the matrix traces by Newton’s recurrence

$$dh_d(P) = \sum_{r=1}^d \text{tr}(P^r) h_{d-r}(P), \quad h_0 = 1.$$

It averages $\prod_j h_{\tau_j}(P)$ over the group. Independently, it extracts the same coefficient from the cycle factors in (1) and counts the orbits in (2) by explicit set traversal. Finally, it evaluates (3) or (5) for every required power and checks that (4) reproduces every actual cycle count.

4 Exact results

The entries list the common value in the order $\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1)$.

Action	$ G $	$ X $	common coefficient vector
$\text{Aut } J(4, 2)$ (including T)	48	6	(1, 3, 3, 5, 8, 11)
S_5 on 2-subsets	120	10	(1, 3, 3, 7, 11, 15)
S_5 on 1-subsets	120	5	(1, 2, 2, 3, 4, 5)

All 18 triple comparisons are exact, and all fixed-point reconstructions pass.

At $(t_1, t_2) = (0.037, 0.061)$ the direct determinant average and cycle product for $\text{Aut } J(4, 2)$ coincide at

$$1.1237420509116285 \quad (\text{reported error } 0).$$

The pair values also give $[m_{(2)}] = [m_{(1,1)}] = \min(k, v - k) + 1$, as predicted.

5 Reproduction

From the repository root run

```
cd for_this_guy/association_scheme_permutation_representations
marimo run johnson/notebook.py
```

or run the family module directly with Python. The notebook displays every matrix/cycle/orbit comparison rather than only the summary above.